

Converting Binary to Decimal

To convert an 8-bit binary number to decimal, use the **place-value method**, where each binary digit is multiplied by its corresponding power of 2, starting from the rightmost digit, and the results are summed up.

An 8-bit number has place values ranging from 2^0 to 2^7 . (1 to 128)

Step-by-Step Conversion

1. **Write down** the 8-bit binary number.
2. **Align** the place values (128, 64, 32, 16, 8, 4, 2, 1) above their corresponding binary digits from left to right.
3. **Multiply** each binary digit (bit) by its corresponding place value. (This means you only consider the place values where the binary digit is '1').
4. **Add** the resulting decimal values together to get the final decimal number.

Example: **10110010**

128	64	32	16	8	4	2	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	0	1	1	0	0	1	0
128	0	32	16	0	0	2	0

Calculation: $(128+0+32+16+0+0+2+0=178)$

Answer : 178

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